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# Paleomagnetism and $^{40}Ar/^{39}Ar$ Chronology of Lavas from Meseta del Lago Buenos Aires, Patagonia

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[1] As part of a larger project investigating paleosecular variation in South America, 36 paleomagnetic and geochronology sites were sampled on Meseta del Lago Buenos Aires (47°S, 289°E), 200 km east of the active plate boundary in southern Patagonia. Basaltic lavas sampled range from late Miocene to late Pleistocene yet most of the individual lavas are younger than 3.3 Ma. Thirty-eight isochron ages determined via $^{40}Ar/^{39}Ar$ incremental-heating experiments frame a detailed stratigraphy. The isochron ages range from 66.9 ± 4.1 ka for scoria cones and a youthful lava flow in the Rio Pinturas valley to 10.12 ± 1.35 Ma for thick plateau-forming lavas exposed along the SE edge of the Meseta. The last 3.3 myr is characterized by seven episodes of volcanism at ca. 3.2–3.0 Ma, 2.4 Ma, 1.7 Ma, 1.35 Ma, 1.0 Ma, 750 ka, 430–330 ka, and <110 ka. The bulk of lavas forming the surface of the Meseta erupted in the last 1.7 my. All sites have stable magnetization, and after step demagnetization using either thermal or alternating field techniques, yield characteristic directions held by magnetite and/or titanomagnetite. Ten sites have distinct transitional directions (defined by pole latitudes <55°), and associated ages indicate possible connections to known reversals within the Matuyama Chron, including the onset of the Jaramillo subchron (1.016 ± 0.01 Ma), the Cobb Mountain subchron (1.25 ± 0.03 Ma), the Ontong-Java 1 event (1.37 ± 0.03 Ma), and the termination of the Olduvai subchron (1.72 ± 0.02 Ma). Remaining sites are divided into normal (14 sites) and reversed polarity (12 sites). At 95% confidence mean normal and reversed directions overlap. The mean direction for 26 sites is I = −63.0°, D = 3.4°, $\alpha_{95}$ = 5.4°, which is indistinguishable from the expected geocentric axial dipole. Paleosecular variation, measured by the dispersion of virtual geomagnetic poles about the rotation axis, is higher than expected at 20.0°. Model G, using data for the past 5 Myr, predicts a dispersion of 17°. This discrepancy may be due to true dispersion of the field in southern South America or it may be an artifact of inadequate sampling.

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**Theme:** Geomagnetic Field Behavior over the Past 5 Myr
**Guest Editors:** Cathy Constable and Catherine Johnson

## 1. Introduction

[2] The magnetic field of the Earth is known to vary over time periods ranging from milliseconds to millions of years. In an attempt to fully understand aspects of the time-averaged field (TAF), considerable effort has gone into studying the paleosecular variation of the field over millions of years. Research in this area falls into several categories including: detailed studies of magnetic field behavior as presented by paleomagnetic field directions at one location on the Earth's surface, modeling experiments concerned with the global representation of the magnetic field, and theoretical studies of field behavior including expected normal variations.

[3] Paleosecular variation studies on lava flows (PSVL) use the instantaneous nature of magnetization in lava flows to sample the Earth's field at discrete positions over long time periods. From the early 1960s onward models have been developed and refined to investigate the variation of PSV over latitude [*Merrill et al.*, 1996]. Recently, a number of studies have investigated TAF by applying global data sets of PSVL to field models [*Johnson and Constable*, 1996, 1997, 1998; *McElhinny and McFadden*, 1997]. *McElhinny and McFadden* [1997] suggest that many older individual studies are inadequate for current needs, although remeasurement of some of those earlier studies have verified the original results [*Brown*, 2002]. In any event, it is necessary to produce not only accurate results but also to increase geographic coverage to constrain field behavior globally. This is especially germane to the Southern Hemisphere, where fewer studies of high quality exist.

[4] In this paper we present paleomagnetic and geochronologic results from Meseta del Lago Buenos Aires, Patagonia, a Miocene - Pleistocene volcanic field at 47°S in the southern part of South America. This study provides important new data on the behavior of the magnetic field over the last few million years from the Southern Hemisphere for comparison to other studies in the Southern Hemisphere, to results from similar latitudes in the Northern Hemisphere, and to provide well-dated and determined magnetic data for incorporation into global models of the field. All radioisotopic ages reported or discussed here are relative to standard values of *Renne et al.* [1998] and the ages of geomagnetic chrons and their boundaries [e.g., *Cande and Kent*, 1995; *Berggren et al.*, 1995] have been adjusted accordingly.

## 2. Geologic Setting

[5] Meseta del Lago Buenos Aires (MLBA) is a basaltic plateau that has a total area of ~5000 km$^2$ and is located in the northwestern corner of the Santa Cruz Province, Argentina, between 46.25°–47.25°S and 288.25°–289.50°E (Figures 1 and 2). The MLBA is a quasi-planar plateau with an ambient elevation of ca. 1400 masl, that is dotted with more than 100 scoria cone vents, many of which attain peak elevations of over 1600 mal [*Baker et al.*, 1981; *Mercer and Sutter*, 1982; *Singer et al.*, 2004] (Figure 2). The Monte Zeballos volcanic complex near the southwestern corner of MLBA rises to 2700 masl (Figure 2). The MLBA comprises Cretaceous and Tertiary sediments capped by 10–30 m thick, columnar-jointed, laterally extensive tholeiitic lavas that we will show below were erupted between ca. 11 and 7 Ma, and which in turn are perforated and overlain by voluminous Pliocene to Quaternary alkaline lavas and intrusions (Figure 2) [*Baker et al.*, 1981; *Gorring et al.*, 2003]. These rocks rest unconformably over Jurassic rhyolites and upper Proterozoic to Devonian low-grade metamorphic basement [*Hervé et al.*, 1981; *Pankhurst et al.*, 1998].

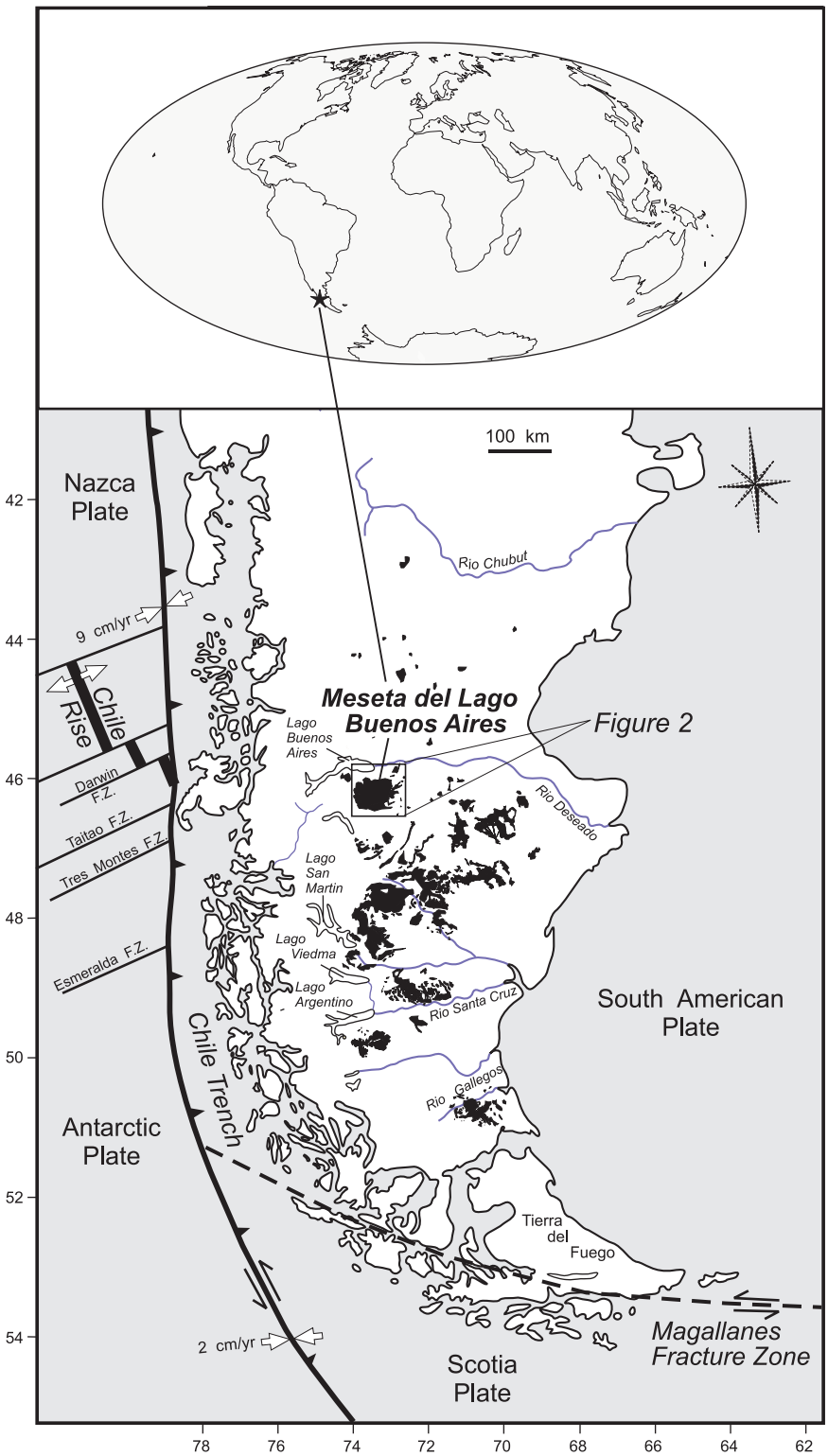



**Figure 1.** Location of the Meseta del Lago Buenos Aires region (highlighted square) in southern Patagonia, Argentina. The Meseta lies some 250 kms east of the triple point between the Nazca, Antarctic and South American plates. [After *Singer et al.*, 2004; *Gorring et al.*, 2003].

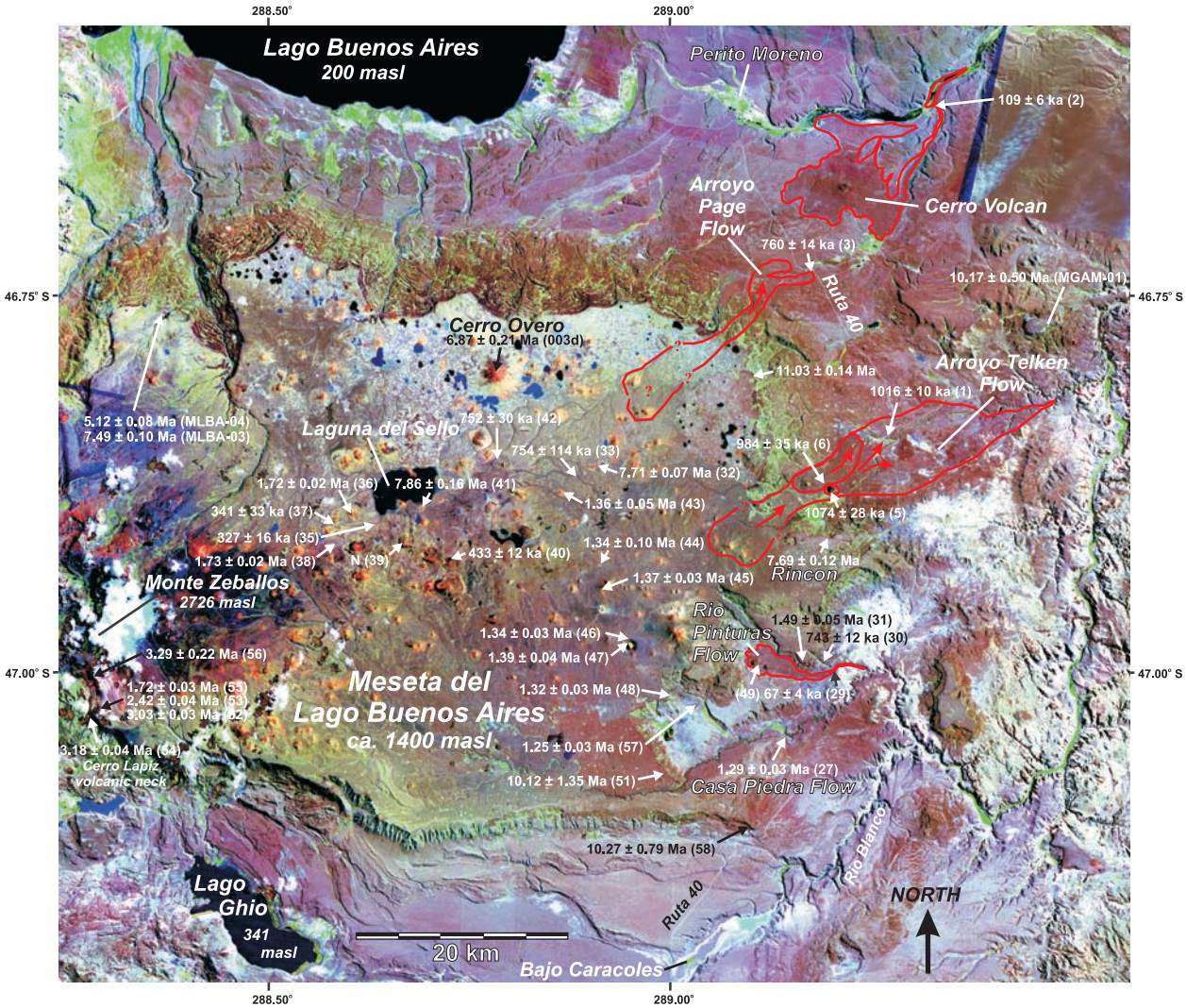



**Figure 2.** Landsat mosaic of the Meseta del Lago Buenos Aires region between the villages of Perito Moreno in the north and Bajo Caracoles in the south. The Argentine-Chilean border lies just west of the area shown. The extent of individual lavas that erupted or flowed away from the Meseta surface are outlined in red. Numbers give the $^{40}Ar/^{39}Ar$ ages and 2σ uncertainties associated with each dated lava, PAT (paleomagnetic sites) numbers are in parentheses. In addition to the new ages in Table 2, three $^{40}Ar/^{39}Ar$ ages from *Ton-That et al.* [1999] are shown, including 10.17 Ma Meseta Gambarana (MGAM-1), an erosional remnant to the northeast of the main plateau, and two lavas from the northwest corner of the Meseta (MLBA-03, and -04) corresponding to the section that was K-Ar dated by *Mercer and Sutter* [1982].

[6] The MLBA is part of a large province of Late Miocene to Quaternary basaltic magmatism that occurs east pf the Andes between 34°S to 52°S [e.g., *Stern et al.*, 1990; *Ramos and Kay*, 1992; *Gorring and Kay*, 2001]. Throughout southern Patagonia (46°–52°S) this magmatism is thought to originate as a consequence of diachronous subduction of segments of the Chile Rise spreading center (Figure 1) and the opening of lithospheric slab windows through which asthenospheric mantle upwelled and partially melted [*Ramos and Kay*, 1992; *Gorring et al.*, 1997, 2003]. Additional details on the regional geology and tectonics can be found in *Ramos and Kay* [1992], *Gorring et al.* [1997, 2003] and *Singer et al.* [2004].

## 3. Field Sampling

[7] MLBA lies in the cool rain shadow of the Andes, thus the climate is arid, rates of weathering and

erosion are relatively low, and remarkably fresh exposures of numerous lava flows are preserved. Sites for paleomagnetic study were sampled in 1998 and 2001 using a gasoline powered drill, with 6 to 10 samples drilled over several square meters of exposed lava at each site. Sample orientations were collected using sun and magnetic compasses and locations established by GPS and plotting on regional topographic maps. Samples for $^{40}Ar/^{39}Ar$ dating were either drilled or chiseled from outcrops at the same locations as where paleomagnetic cores were taken. To the north and east of MLBA access was via national highway 40, whereas the rugged interior, the southwest margin of the plateau, and the steep western side are accessible only via four-wheel drive vehicle and traverses on foot (Figure 2).

[8] A total of 36 sites were drilled for paleomagnetic studies; all sites but two have associated geochronologic samples available for study. The two undated sites are from flows that are in direct stratigraphic contact with dated flows. Sites are concentrated in the plateau-capping flows, individual volcanic cones, and flows that spilled over the plateau into the broad valley to the east, although a small number of sites were collected in the older plateau-forming flows.

## 4. Paleomagnetic Methods

[9] Magnetic data was processed in the Paleomagnetism Laboratory at the University of Massachusetts. Samples were measured on a 2G cryogenic magnetometer, model 755R, with alternating field and thermal demagnetization done using a Molspin AF demagnetizer and an ASC thermal demagnetizer, respectively. Susceptibility measurements were done using a Sapphire susceptibility meter.

[10] Paleomagnetic samples were subjected to detailed demagnetization using both alternating field (AF) and thermal techniques. AF demagnetization was performed in steps from 0 to 100 mT, with 10 to 12 steps per sample. Thermal demagnetization, done at 10 temperature steps from room temperature to 600°C, was used for a minimum of two samples from each flow. Characteristic directions were determined for both demagnetization methods using line-fitting techniques [*Kirschvink*, 1980] using at least 4 points for each calculation and obtaining maximum angles of deviation of less than 5°.

## 5. The $^{40}Ar/^{39}Ar$ Methods

[11] At the University of Wisconsin-Madison incremental heating experiments used a metal resistance furnace to degas ca. 25 mg aliquots of holocrystalline groundmass in 5 to 11 steps between 600° and 1375°C. Complete degassing of the feldspar separate from sample PAT-56 required 1675°C. Procedures for preparing and irradiating samples, extracting gas from samples and fluence monitors, correcting for blanks, mass spectrometry, and estimating analytical uncertainties are detailed in *Singer et al.* [2002] and *Singer and Brown* [2002]. Results obtained using identical procedures on several of the lava flows including PAT-2, PAT-3, PAT-5, and PAT-6, are fully reported in *Singer et al.* [2004], whereas *Ackert et al.* [2003] summarize the multiple experiments on groundmass from samples of sites PAT-49 and PAT-29 on the Rio Pinturas basalt flow.

[12] The $^{40}Ar/^{39}Ar$ method requires that the age of a sample be calculated relative to a mineral standard which has been previously dated, usually by conventional K-Ar techniques. The monitor mineral used here was sanidine from the Alder Creek rhyolite (ACs). *Turrin et al.* [1994] measured the age of ACs at 1.186 Ma relative to 27.84 Ma sanidine from the Fish Canyon tuff. In light of recent intercalibration of these two $^{40}Ar/^{39}Ar$ standards relative to 98.79 ± 0.96 Ma GA-1550 biotite, we have adopted an age of 1.194 Ma for ACs [*Renne et al.*, 1998]; however, a consensus regarding the age of the GA-1550 standard has not yet been reached [*Lanphere and Dalrymple*, 2000]. We note that adopting an age of 1.186 Ma for the ACs standard, equivalent to assuming an age for Fish Canyon tuff sanidine of 27.84 Ma that is compatible with the timescales of *Cande and Kent* [1995] and *Berggren et al.* [1995] would shift the ages reported here about 1% younger, but would not alter any of our conclusions. Conversion of our ages to make them consistent with other values for standard ages, e.g., *Villeneuve et al.* [2000], can be done simply using the equation in *Dalrymple et al.*

[1993]. Samples were irradiated for one hour adjacent to ACs monitors in evacuated quartz vials at the Oregon State University Triga reactor in the Cadmium-Lined In-Core Irradiation Tube (CLICIT). Corrections for undesirable nucleogenic reactions on $^{40}K$ and $^{40}Ca$ are: $[^{40}Ar/^{39}Ar]_K$ = 0.00086; $[^{36}Ar/^{37}Ar]_{Ca}$ = 0.000264; $[^{39}Ar/^{37}Ar]_{Ca}$ = 0.000673 [*Wijbrans et al.*, 1995].

[13] Inverse-variance weighted mean plateau ages and uncertainties are calculated according to *Taylor* [1982]. Precision estimates for monitors, based on 5–6 measurements each suggest that the uncertainty in *J*, the neutron fluence parameter, was typically 0.6% (±2σ); this uncertainty was propagated into the final plateau and isochron age for each analysis, but contributes <0.1% to the total uncertainty in these age estimates. Ages were calculated using the decay constants of *Steiger and Jäger* [1977] and are reported with ±2σ analytical uncertainties.

[14] Criteria used to determine ages were: (1) age plateaus are defined by at least three contiguous steps all concordant in age at the 95% confidence level and comprising >50% of the $^{39}Ar$ released, (2) a well-defined isochron exists for the plateau points as defined by the F-variate statistic *SUMS*/($N - 2$) [*York*, 1969], and (3) the isochron ages [*York*, 1969] are preferred over the weighted mean plateau ages because they combine estimates of analytical precision plus internal disturbance of the sample (scatter about the isochron) and they make no assumption about the trapped argon component. Where multiple subsamples were measured from a lava, the isochron combining all the plateau steps gives our best estimate of the age of the flow [*Singer et al.*, 2002, 2004].

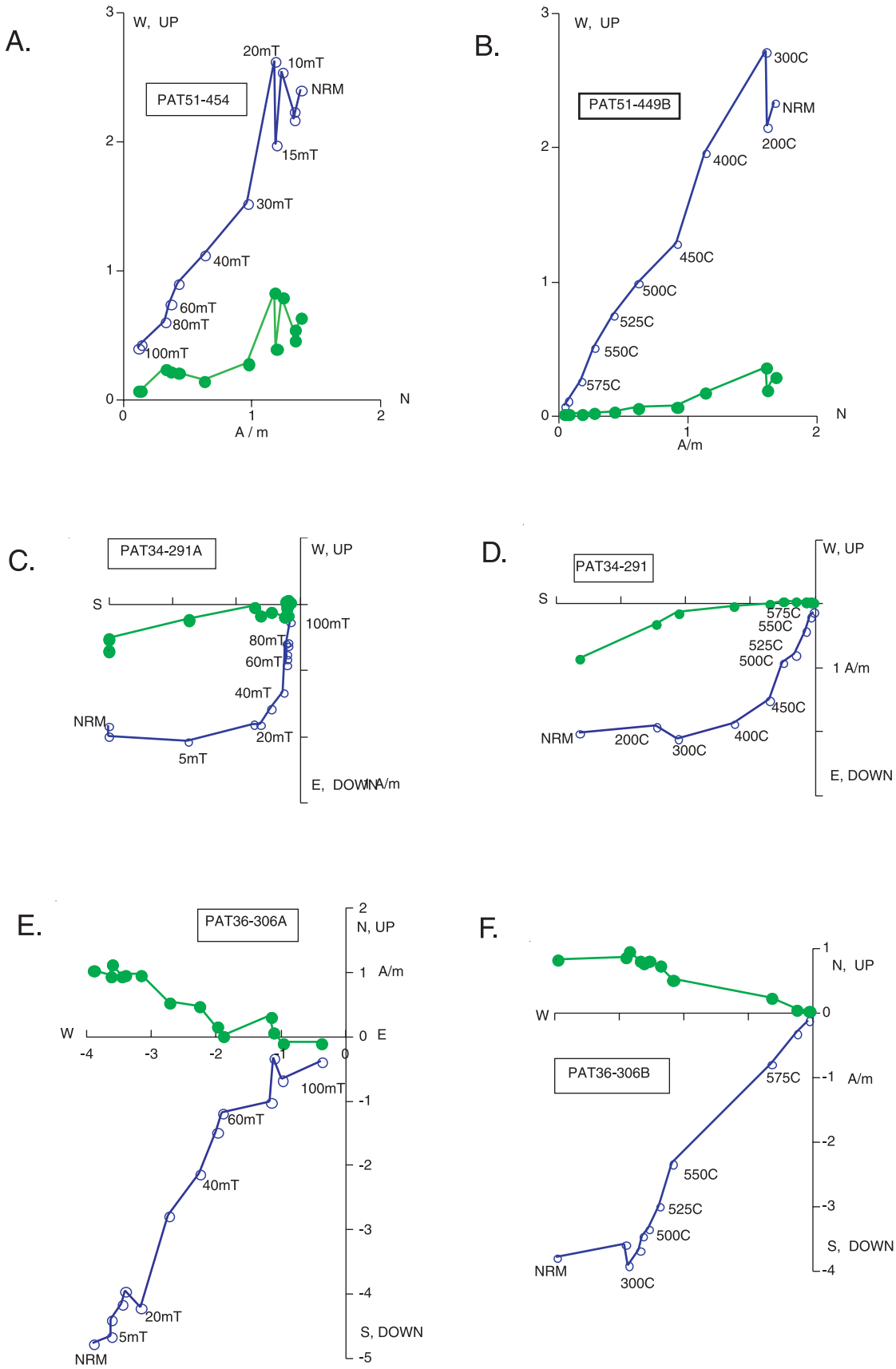
## 6. Paleomagnetic Results

[15] Natural remanent magnetization (NRM) and demagnetization behavior was measured on a total of 274 samples from 36 sites on or near Meseta del Lago Buenos Aires. Both alternating field and thermal demagnetization techniques were used on samples from the same site with consistent agreement between the two methods (Figure 3). AF demagnetization on normal flows removed only minor overprints at low levels (Figure 3a) while reversed flows showed larger overprints, but ones that were removed by fields of 20 to 40 mT (Figure 3c). Thermal demagnetization on most samples showed similar responses with small overprints removed from the normal flows by 300°C (Figure 3b) and large overprints removed from the reversed flows at levels of 400 to 500°C (Figure 3d). Samples from sites having transitional directions also showed small but varying amounts of overprinting and then straight-line decay to the origin with both AF and thermal techniques (Figure 3e and 3f).

[16] Susceptibility measurements made on each core and averaged for each site are shown in the histogram in Figure 4a. Values range from $0.55 \times 10^{-2}$ to $9.51 \times 10^{-2}$ SI, with a mean of $2.23 \times 10^{-2}$ SI. NRM values before demagnetization range from 1.08 A/m to 8.00A/m (Figure 4b), with a mean of 4.03 A/m. Sites show no indication of high magnetizations (>20 A/m) associated with lightning strikes [*Cox*, 1961; *Tauxe et al.*, 2003] nor do any show exceedingly low magnetizations (<1 A/m) as might be associated with times of low field strength such as excursions or reversals.

[17] Inclinations and declinations for all the flows, along with associated statistics and virtual geomagnetic poles (VGP) are given in Table 1 and plotted in Figure 5a. Most flows show excellent within site statistics, precision parameter (k) and 95% circles of confidence ($\alpha_{95}$) of *Fisher* [1953]. A histogram of $\alpha_{95}$ (Figure 4c) indicates that only 6 flows have $\alpha_{95}$ values greater than 5.5°, with a mean value of all flows of 4.9° and a range of 2.6° to 13.9°.

**Figure 3.** Vector end-point diagrams showing typical demagnetization behavior of normal, reversed, and transitional samples from Meseta del lago Buenos Aires. Left-hand sides are alternating field demagnetization, right-hand side are thermal demagnetization. (a, b) Normally magnetized lava flow, with alternating field and thermal demagnetization on two different samples. (c, d) Reversely magnetized lava flows showing large overprint removed in both alternating field and thermal demagnetization on two specimens from the same core. (e, f) Examples of two specimens from the same core from a transitional site. Open circles on blue (dark) lines are projections onto the vertical plane; solid circles on green (light) lines are projections on the horizontal plane.


A.
W, UP
PAT51-454
20mT
10mT
NRM
15mT
30mT
40mT
60mT
80mT
100mT
N
A / m
B.
W, UP
PAT51-449B
300C
NRM
200C
400C
450C
500C
525C
550C
575C
N
A/m
C.
PAT34-291A
W, UP
S
100mT
80mT
60mT
40mT
NRM
20mT
5mT
E, DOWN
A/m
D.
PAT34-291
W, UP
S
575C
550C
525C
500C
1 A/m
450C
NRM
200C
300C
400C
E, DOWN
E.
PAT36-306A
N, UP
1 A/m
W
0 E
100mT
60mT
40mT
20mT
5mT
NRM
S, DOWN
F.
N, UP
W
PAT36-306B
575C
-1 A/m
550C
525C
500C
NRM
300C
S, DOWN

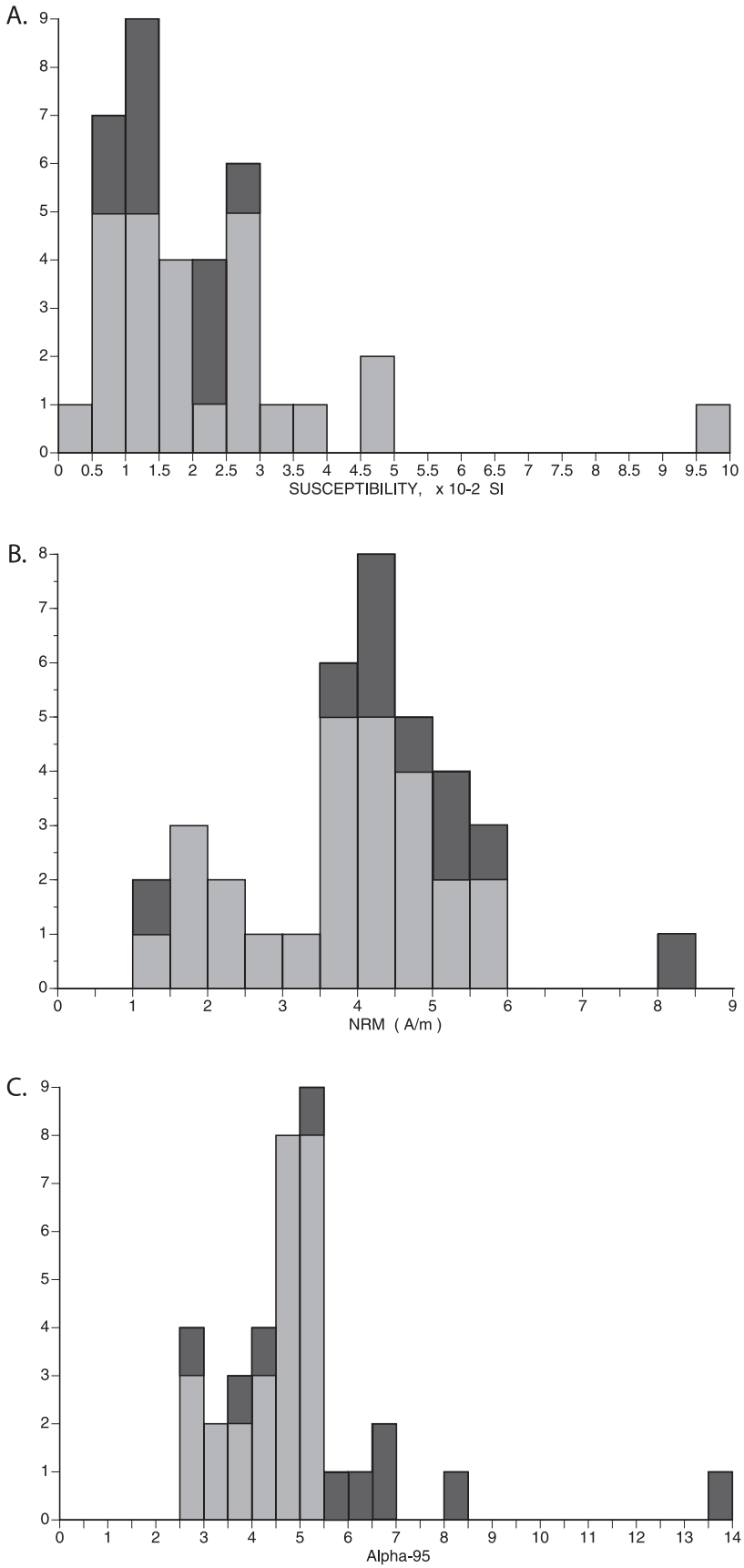

A.
SUSCEPTIBILITY, x 10-2 SI
B.
NRM ( A/m )
C.
Alpha-95

**Table 1.** Summary of Site Mean Paleomagnetic Data on Lavas From Meseta Lago Buenos Aires[a]

| SITE | Site Lat. | Site Long. | Age (Ma) ± 2sigma | N | Polarity | Inc | Dec | a-95 | K | Pole Lat. | Pole Long. |
|---|---|---|---|---|---|---|---|---|---|---|---|
| PAT 1 | −46.87 | 289.26 | 1.016 ± 0.010 | 6/8 | T | −47.7 | 183.6 | 5.6 | 145 | −14.0 | 112.5 |
| PAT 2 | −46.70 | 289.25 | 0.109 ± 0.006 | 8/8 | N | −73.0 | 15.7 | 3.6 | 245 | 74.9 | 76.4 |
| PAT 3 | −46.70 | 289.16 | 0.760 ± 0.014 | 9/9 | N | −60.9 | 9.8 | 2.9 | 327 | 81.5 | 348.6 |
| PAT 5 | −46.90 | 289.30 | 1.074 ± 0.028 | 5/5 | N | −66.9 | 326.2 | 4.7 | 264 | 67.5 | 180.1 |
| PAT 6 | −46.92 | 289.20 | 0.984 ± 0.035 | 8/8 | T | −84.8 | 317.3 | 4.3 | 165 | 54.0 | 121.2 |
| PAT 27 | −47.13 | 289.14 | 1.290 ± 0.030 | 10/10 | T | 84.6 | 34.6 | 6.6 | 54 | −38.1 | 296.8 |
| PAT 28 | −47.13 | 289.14 | no date | 6/7 | T | 77.6 | 75.7 | 6.4 | 112 | −37.1 | 318.4 |
| PAT 29 | −47.10 | 289.20 | 0.070 ± 0.024 | 8/8 | N | −50.1 | 9.7 | 2.7 | 425 | 72.2 | 317.5 |
| PAT 30 | −47.06 | 289.18 | 0.743 ± 0.012 | 5/7 | N | −70.6 | 20.1 | 4.8 | 194 | 75.2 | 58.1 |
| PAT 31 | −47.06 | 289.18 | 1.490 ± 0.050 | 8/9 | R | 70.9 | 167.0 | 5.2 | 113 | −78.5 | 329.4 |
| PAT 32 | −46.90 | 288.90 | 7.710 ± 0.070 | 9/9 | R | 64.3 | 218.4 | 3.6 | 207 | −63.9 | 211.3 |
| PAT 33 | −46.92 | 288.90 | 0.754 ± 0.114 | 9/10 | N | −73.6 | 5.4 | 4.7 | 121 | 77.0 | 96.5 |
| PAT 34 | −46.90 | 288.81 | 1.143 ± 0.064 | 6/9 | R | 77.8 | 153.2 | 4.2 | 253 | −65.8 | 314.7 |
| PAT 35 | −46.95 | 288.64 | 0.327 ± 0.016 | 10/10 | N | −75.5 | 33.3 | 3.0 | 261 | 65.7 | 70.9 |
| PAT 36 | −46.91 | 288.60 | 1.720 ± 0.020 | 10/10 | T | 50.1 | 279.5 | 2.6 | 341 | −16.1 | 226.8 |
| PAT 37 | −46.96 | 288.57 | 0.341 ± 0.033 | 7/9 | T | −80.7 | 55.5 | 3.5 | 294 | 54.6 | 82.3 |
| PAT 38 | −46.97 | 288.58 | 1.730 ± 0.020 | 5/9 | T | 47.2 | 321.9 | 13.9 | 32 | 7.2 | 255.4 |
| PAT 39 | −46.97 | 288.67 | 0.336 ± 0.012 | 3/5 | N | −80.0 | 339.4 | 4.4 | 788 | 64.4 | 124.4 |
| PAT 40 | −46.97 | 288.68 | 0.433 ± 0.012 | 6/6 | N | −48.8 | 5.0 | 2.8 | 459 | 72.3 | 303.0 |
| PAT 41 | −46.90 | 288.70 | 7.860 ± 0.160 | 9/9 | T | −16.6 | 112.0 | 5.0 | 108 | −8.4 | 40.8 |
| PAT 42 | −46.90 | 288.79 | 0.752 ± 0.030 | 10/10 | N | −70.2 | 50.8 | 4.5 | 119 | 57.6 | 51.0 |
| PAT 43 | −46.90 | 288.91 | 1.360 ± 0.050 | 6/10 | R | 67.8 | 165.3 | 4.5 | 218 | −79.6 | 351.8 |
| PAT 44 | −46.95 | 288.93 | 1.341 ± 0.103 | 10/10 | R | 56.9 | 169.6 | 4.9 | 100 | −77.9 | 65.8 |
| PAT 45 | −47.00 | 289.10 | 1.370 ± 0.030 | 8/8 | R | 61.6 | 187.5 | 5.1 | 118 | −83.2 | 162.8 |
| PAT 46 | −47.05 | 288.95 | 1.390 ± 0.040 | 8/8 | R | 59.9 | 177.5 | 5.2 | 115 | −83.5 | 92.2 |
| PAT 47 | −47.05 | 288.95 | 1.340 ± 0.030 | 8/8 | R | 73.1 | 184.8 | 5.3 | 132 | −78.1 | 276.8 |
| PAT 48 | −47.10 | 289.00 | 1.315 ± 0.029 | 9/9 | T | 81.6 | 27.8 | 8.2 | 41 | −32.1 | 298.0 |
| PAT 49 | −47.09 | 289.16 | 0.067 ± 0.004 | 10/10 | N | −64.0 | 16.1 | 5.3 | 84 | 78.8 | 17.8 |
| PAT 50 | −47.09 | 289.16 | no date | 8/9 | N | −43.9 | 356.2 | 4.6 | 144 | 68.4 | 279.9 |
| PAT 51 | −47.17 | 289.02 | 10.120 ± 1.350 | 10/10 | N | −62.4 | 334.2 | 3.2 | 235 | 71.6 | 200.3 |
| PAT 52 | −47.12 | 288.28 | 3.030 ± 0.030 | 5/6 | R | 38.2 | 200.0 | 4.3 | 316 | −59.7 | 147.4 |
| PAT 53 | −47.12 | 288.28 | 2.422 ± 0.035 | 7/7 | R | 49.1 | 174.7 | 5.0 | 148 | −72.4 | 92.9 |
| PAT 54 | −47.12 | 288.27 | 3.180 ± 0.040 | 5/6 | N | −47.5 | 27.4 | 5.2 | 221 | 61.8 | 347.1 |
| PAT 55 | −47.10 | 288.25 | 1.709 ± 0.026 | 5/6 | R | 49.5 | 146.9 | 5.4 | 204 | −59.6 | 39.7 |
| PAT 56 | −47.00 | 288.24 | 3.290 ± 0.220 | 7/7 | R | 43.5 | 186.5 | 4.7 | 167 | −67.8 | 123.9 |
| PAT 57 | −47.11 | 289.02 | 1.250 ± 0.031 | 10/10 | T | 68.4 | 32.2 | 6.5 | 57 | −12.5 | 308.8 |

[a] N, number of samples used in site calculations; Polarity, direction of the field; N, normal; R, reversed; T, transitional, selection described in text; a-95; and K, radius of 95% cone of confidence about the mean direction and precision parameter [*Fisher*, 1953].

[18] In recent years strict criteria have been suggested for determining sites suitable to represent the time averaged field [*Johnson et al.*, 1998; *Tauxe et al.*, 2000]. One important criterion is that $\alpha_{95}$ values must be <5° (rounded to the nearest integer) For this data set that would exclude the 6 sites with $\alpha_{95} > 5.5°$. This criterion is considerably stricter than earlier constraints on $\alpha_{95}$, when 20° was often used as a cut-off [*McElhinny and McFadden*, 1997]. Another criterion usually applied is to omit sites with low latitude poles, the often-called "transitional" directions. A common latitude cut-off angle is 45°, although some authors have used smaller values [*Johnson and Constable*, 1996], larger values [*Constable and Johnson*, 1999] or argued for a variable cut-off size depending on the angular dispersion [*Vandamme*, 1994]. For this latitude the *Vandamme* [1994] method would require a cut-off angle of around 35° [*McElhinny and McFadden*, 1997], noting that in

**Figure 4.** Histograms of site mean data from MLBA. (a) Mean magnetic susceptibility of each flow. (b) Natural remanent magnetization, before demagnetization, for each flow. (c) Alpha-95, radius of cone of confidence at 95%, in degrees, for each flow. N = 36 for each histogram. Dark shaded flows represent sites with transitional directions; see text for criteria.

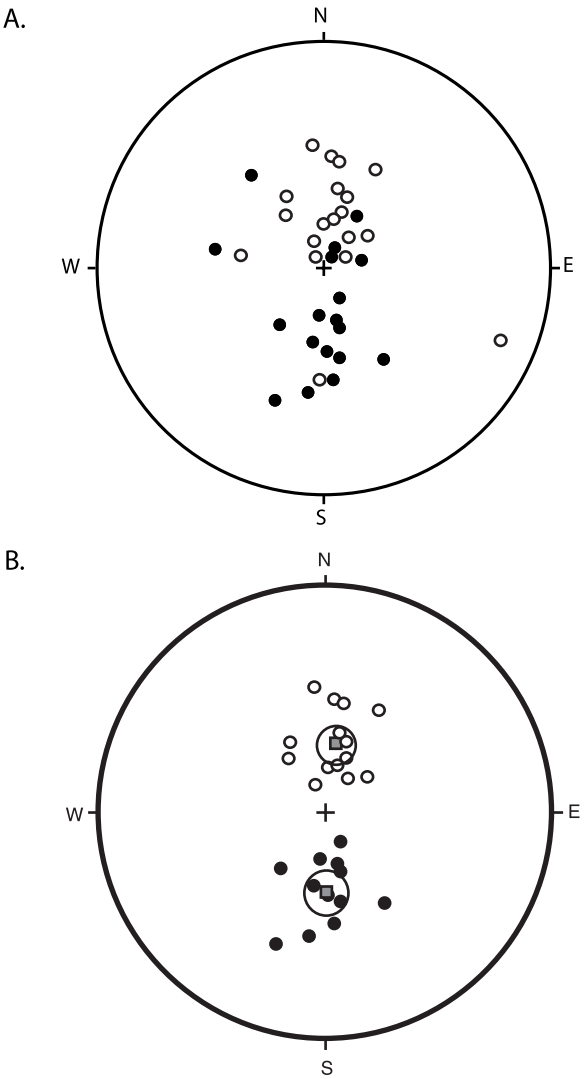



**Figure 5.** Equal area nets of inclination and declination, determined from principal component analysis, for paleomagnetic sites from Meseta del Lago Buenos Aires. (a) All sites measured. (b) Selected sites, omitting ones with VGP latitudes <55°. Mean directions for the separate normal and reversed populations are shown with their 95% circles of confidence. Open circles are upward pointing (negative) inclinations, solid circles are downward pointing (positive) inclinations.

this case the angle is referring to the co-latitude and not the actual pole latitude.

[19] When the two criteria are applied to our data set from Meseta del Lago Buenos Aires 10 sites are removed from the population, the six sites with $\alpha_{95}$ above 5.5°, which all happen to also have VGP latitudes <55°, and four additional sites, all with small $\alpha_{95}$ values but pole latitudes <55°. Details of these shallow latitude sites, collectively called transitional, are given in Table 3. A total of 26 sites fulfill the criteria to represent the time averaged field, with 14 sites having normal polarity and 12 sites having reversed polarity, and are referred to as Selection I. The directions of this subset are plotted in Figure 5b along with the mean directions for both normal and reversed polarity sites.

[20] The normal and reversed means, with the reversed direction inverted through the origin, overlap, as can be seen in Figure 6. The geocentric axial dipole (GAD) field for this latitude lies directly between the normal and inverted reversed directions. Also shown is the direction of the IGRF2000 field for this location, which is considerably shallower than the GAD.

[21] Recently *Tauxe et al.* [2003] have suggested that the precision parameter, k, is a better determinant of data quality, and that at least 5 samples per site are needed to accurately determine k. If these criteria are applied to the Meseta del Lago Buenos paleomagnetic sites, six sites are eliminated (five

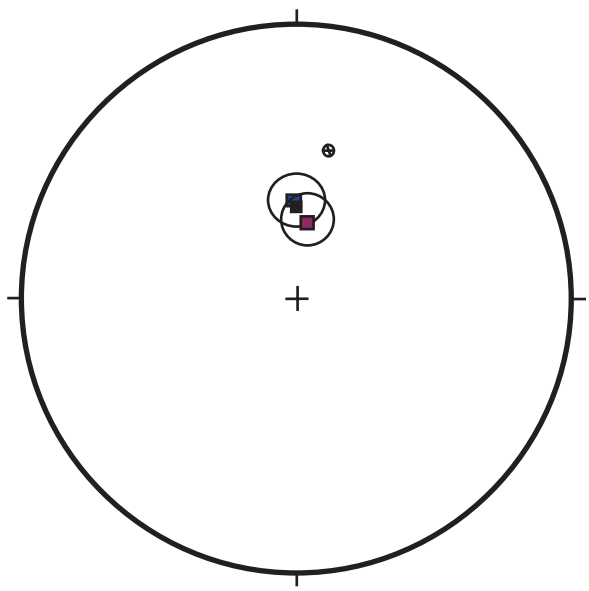


**Figure 6.** Equal-area net of mean directions from normal and reversed populations at Meseta del Lago Buenos Aires. Reversed direction is inverted to appear in the northern hemisphere. Both directions are shown with circles of 95% confidence. Small black square, between normal and reversed means, is position of the geocentric axial dipole field for 47°S. Small x in circle is present field at MLBA from IGRF-2000.

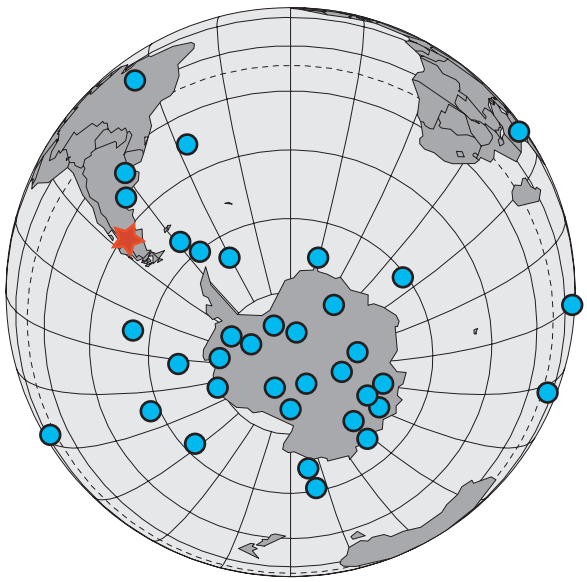


**Figure 7.** South polar projection with all VGPs from Meseta del Lago Buenos Aires plotted. Location of MLBA in southern South America shown by star.

with k values <100 and one with N < 5). This data set, referred to as Selection II, eliminates four of the same transitional flows as in Selection I and two additional flows not rejected previously.

[22] Virtual geomagnetic pole positions for all the sites from MLBA are displayed in the Southern Hemisphere (Figure 7). Statistics for the mean poles from the total population and various subsets are listed in Table 4.

## 7. The $^{40}Ar/^{39}Ar$ Results

[23] Twenty four of the 38 lavas in Table 2 yielded remarkably concordant age spectra with >95% of the gas released defining the plateau ages (Figure 8). In contrast, only seven of these lavas yielded age spectra where <75% of the gas defined the age plateau, yet even these mildly discordant experiments yielded plateau and isochron ages that met the criteria outlined above (Table 2). None of the 38 isochrons yielded a $^{40}Ar/^{36}Ar_i$ value significantly different than 295.5 indicating that excess argon is not present in detectable levels that might bias the results (Figure 8). Moreover the age spectra and isochrons do not suggest that inherited or xenocrystic argon is of concern. Accordingly, we interpret the isochron ages from the 38 samples as the best estimate of time elapsed since these lavas erupted, cooled through their Curie temperatures, and acquired their thermoremanent magnetization direction.

[24] The basaltic lavas range in age from an 11.03 ± 0.14 Ma colonnade at the base of the Meseta-capping flow sequence south of Estancia Telken (Figure 2) to the 66 ± 3 ka Rio Pinturas pahoehoe flow that erupted from vents defined by two small scoria cones near the head of the colorful Rio Pinturas, one km from older eroded lavas that define the eastern escarpment of MLBA (Figure 2). Statistical analysis of the distribution of $^{40}Ar/^{39}Ar$ ages, including three lavas previously dated by *Ton-That et al.* [1999] (Figure 2), reveals that basaltic volcanism was episodic rather than continuous over the last 11 million years (Figure 9). The reconnaissance nature of this study prevents extrapolating erupted volumes solely from the $^{40}Ar/^{39}Ar$ measurements, however, based on our field observations and limited mapping, this approach does yield a first-order impression of how MLBA formed. The ages define at least ten periods of peak volcanism, including multiple flows at ca. 11–10 Ma, 7.3–7.8 Ma, 3.2–3.0 Ma, 2.4 Ma, 1.7 Ma, 1.35 Ma, 1.0 Ma, 750 ka, 430–330 ka, and 109–66 ka (Figure 9). In addition, a single >10 m thick basalt flow on the northwest corner of MLBA, previously K-Ar dated by *Mercer and Sutter* [1982], was $^{40}Ar/^{39}Ar$ dated by *Ton-That et al.* [1999] at 5.12 ± 0.008 Ma (Figure 9). The most prominent peaks occurred during the latter part of the Matuyama reversed chron 1.70, 1.35 and 1.0 Ma and in the Brunhes chron 750 ka, 400 ka, and <110 ka (Figure 9). Both the Late Miocene and Late Pliocene-Pleistocene volcanic episodes in this region are consistent with diachronous subduction of segments of the Chile Rise spreading center, opening of a slab window beneath the South American plate, and melting the underlying asthenospheric mantle [*Gorring et al.*, 2003 and references therein] (Figure 1).

[25] Paleomagnetic directions were determined from 34 of the 37 dated lava flows; normal paleodirections were determined from 16 of the lavas, whereas 12 others gave reversed directions, and 7 yielded transitional directions (Table 1).

**Table 2.** Summary of $^{40}Ar/^{39}Ar$ Incremental Heating Experiments on Lavas From Meseta del Lago Buenos Aires[a]

| Paleomag Site # | Field ID # | Unit or Location | Lat. | Long. | K/Ca total | Total fusion Age (Ma) ± 2σ | Increments used, °C | Age Spectrum | | | | Isochron Analysis | | | Polarity |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | $^{39}Ar$% | Age (Ma) ± 2σ | MSWD | N | SUMS (N-2) | $^{40}Ar/^{36}Ar_i$ ± 2σ | [a]**Age (Ma) ± 2σ** | |
| PAT 49 | MLBA-01-47 | Rio Pinturas basalt flow | −47.09 | 289.16 | 0.43 | 0.072 ± 0.004 | 700–1375 | 99.0 | 0.071 ± 0.003 | 0.93 | 28 of 29 | 0.79 | 296.5 ± 0.9 | [c]**0.067 ± 0.004** | N |
| PAT 29 | MLBA-98-20 | Rio Pinturas basalt flow | −47.10 | 289.20 | 0.35 | 0.075 ± 0.012 | 725–1325 | 100.0 | 0.073 ± 0.001 | 0.79 | 7 of 7 | 0.94 | 295.8 ± 2.7 | [c]**0.070 ± 0.024** | N |
| PAT 2 | CV-98-02 | Cerro Volcan basalt flow | −46.70 | 289.25 | 1.65 | 0.103 ± 0.004 | 800–1200 | 80.0 | 0.108 ± 0.002 | 1.10 | 26 of 38 | 1.12 | 295.1 ± 1.8 | [b]**0.109 ± 0.006** | N |
| PAT 35 | MLBA-01-33 | West Laguna del Sello | −46.95 | 288.64 | 0.31 | 0.322 ± 0.007 | 875–1400 | 100.0 | 0.320 ± 0.007 | 1.53 | 5 of 5 | 1.51 | 293.8 ± 3.3 | **0.327 ± 0.016** | N |
| PAT 39 | MLBA-01-37 | West Laguna del Sello | −46.97 | 288.67 | 0.22 | 0.354 ± 0.014 | 1000–1300 | 68.1 | 0.332 ± 0.007 | 0.82 | 3 of 6 | 0.91 | 294.8 ± 1.5 | **0.336 ± 0.012** | N |
| PAT 37 | MLBA-01-35 | West Laguna del Sello | −46.96 | 288.57 | 0.31 | 0.333 ± 0.019 | 885–1400 | 100.0 | 0.336 ± 0.016 | 0.20 | 4 of 4 | 0.23 | 294.7 ± 4.2 | **0.341 ± 0.033** | T |
| PAT 40 | MLBA-98-10 | Cerro Leon Trachyte | −46.97 | 288.68 | 0.60 | 0.442 ± 0.008 | 700–1350 | 95.0 | 0.435 ± 0.006 | 1.20 | 4 of 6 | 1.67 | 296.0 ± 2.6 | **0.433 ± 0.012** | N |
| PAT 30 | MLBA-01-29 | Older flow north of Rio Pinturas | −47.06 | 289.18 | 0.29 | 0.751 ± 0.001 | 750–1290 | 100.0 | 0.748 ± 0.008 | 0.48 | 6 of 6 | 0.40 | 296.0 ± 1.1 | **0.743 ± 0.012** | N |
| PAT 42 | MLBA-01-39 | East Laguna del Sello | −46.90 | 288.79 | 0.38 | 0.751 ± 0.014 | 725–1250 | 100.0 | 0.752 ± 0.011 | 0.36 | 5 of 5 | 0.47 | 295.5 ± 3.0 | **0.752 ± 0.030** | N |
| PAT 3 | AP-96-01 | Arroyo Page basalt flow | −46.70 | 289.16 | 0.18 | 0.756 ± 0.020 | 550–1150 | 90.2 | 0.764 ± 0.010 | 0.50 | 9 of 13 | 0.60 | 295.9 ± 0.6 | [b]**0.760 ± 0.014** | N |
| PAT 33 | MLBA-01-32 | East Laguna del Sello | −46.92 | 288.90 | 0.35 | 0.764 ± 0.026 | 830–1350 | 100.0 | 0.766 ± 0.022 | 0.46 | 5 of 5 | 0.60 | 296.6 ± 10.2 | **0.754 ± 0.114** | N |
| PAT 6 | AT-98-03 | Estancia La Paloma basalt flow | −46.92 | 289.20 | 1.22 | 0.966 ± 0.018 | 650–1020 | 82.0 | 0.987 ± 0.015 | 1.20 | 8 of 10 | 2.20 | 295.3 ± 6.5 | [b]**0.984 ± 0.035** | T |
| PAT 1 | AT-96-01 | Arroyo Telken | −46.87 | 289.26 | 1.70 | 0.988 ± 0.006 | 575–815 | 61.1 | 1.014 ± 0.002 | 1.60 | 10 of 16 | 1.80 | 293.8 ± 3.8 | [b]**1.016 ± 0.010** | T |
| PAT 5 | AT-98-02 | Lagunita del Rincon | −46.90 | 289.30 | 0.96 | 1.023 ± 0.012 | 650–1040 | 86.2 | 1.047 ± 0.013 | 2.80 | 8 of 10 | 3.20 | 286.6 ± 7.6 | [b]**1.074 ± 0.028** | N |
| PAT 34 | MLBA-98-11 | Cerro Colorado | −46.90 | 288.81 | 0.42 | 1.146 ± 0.028 | 700–1300 | 100.0 | 1.148 ± 0.023 | 0.34 | 6 of 6 | 0.41 | 295.9 ± 4.9 | **1.143 ± 0.064** | R |
| PAT 57 | MLBA-98-18 | Estancia Vizcaino basalt flow | −47.11 | 289.02 | 0.54 | 1.268 ± 0.021 | 700–1325 | 100.0 | 1.264 ± 0.017 | 0.77 | 6 of 6 | 0.64 | 297.7 ± 4.0 | **1.250 ± 0.031** | T |
| PAT 27 | MLBA-01-27 | Casa Piedra, lower basalt flow | −47.13 | 289.14 | 0.40 | 1.290 ± 0.020 | 725–1225 | 100.0 | 1.300 ± 0.010 | 0.68 | 5 of 5 | 0.90 | 295.7 ± 2.6 | **1.290 ± 0.030** | T |

**Table 2.** (continued)

| Paleomag Site # | Field ID # | Unit or Location | Lat. | Long. | K/Ca total | Total fusion Age (Ma) ± 2σ | Increments used, °C | Age Spectrum | | | N | Isochron Analysis | | | Polarity |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | $^{39}$Ar% | Age (Ma) ± 2σ | MSWD | | SUMS (N-2) | $^{40}$Ar/$^{36}$Ar$_i$ ± 2σ | [a]**Age (Ma) ± 2σ** | |
| PAT 48 | MLBA-98-017 | SE MLBA, Estancia Vizcaino | −47.10 | 289.00 | 0.54 | 1.300 ± 0.010 | 700–1300 | 100.0 | 1.310 ± 0.020 | 1.82 | 5 of 5 | 2.33 | 295.0 ± 3.4 | **1.315 ± 0.029** | R |
| PAT 47 | MLBA-98-14 | Laguna Honda crater top | −47.05 | 288.95 | 0.26 | 1.310 ± 0.030 | 900–1350 | 95.0 | 1.340 ± 0.020 | 0.24 | 5 of 7 | 0.32 | 295.5 ± 2.9 | **1.340 ± 0.030** | R |
| PAT 44 | MLBA-01-41 | East Laguna del Sello | −46.95 | 288.93 | 0.29 | 1.446 ± 0.056 | 1165–1375 | 67.1 | 1.410 ± 0.003 | 0.88 | 4 of 9 | 0.30 | 299.0 ± 4.9 | **1.341 ± 0.103** | R |
| PAT 43 | MLBA-01-40 | East Laguna del Sello | −46.90 | 288.91 | 0.37 | 1.340 ± 0.020 | 700–1350 | 100.0 | 1.340 ± 0.030 | 1.58 | 6 of 6 | 1.74 | 292.5 ± 6.7 | **1.360 ± 0.050** | R |
| PAT 45 | MLBA-98-13 | East Laguna del Sello | −47.00 | 289.10 | 0.37 | 1.360 ± 0.030 | 700–1300 | 100.0 | 1.360± 0.030 | 0.42 | 6 of 6 | 0.44 | 294.5 ± 3.7 | **1.370 ± 0.030** | R |
| PAT 46 | MLBA-98-15 | Laguna Honda crater bottom | −47.05 | 288.95 | 0.66 | 1.400 ± 0.020 | 925–1250 | 99.4 | 1.410 ± 0.020 | 0.45 | 4 of 7 | 0.36 | 299.3 ± 9.7 | **1.390 ± 0.040** | R |
| PAT 31 | MLBA-01-30 | Above Rio Pinturas valley | −47.06 | 289.18 | 0.56 | 1.490 ± 0.020 | 875–1260 | 100.0 | 1.490 ± 0.020 | 0.36 | 6 of 6 | 0.45 | 295.5 ± 1.4 | **1.490 ± 0.050** | R |
| PAT 55 | MLBA-01-54 | Cerro Lapiz section top flow | −47.10 | 288.25 | 0.29 | 1.715 ± 0.024 | 700–1300 | 100.0 | 1.714 ± 0.018 | 0.56 | 7 of 7 | 0.61 | 295.9 ± 1.6 | **1.709 ± 0.025** | R |
| PAT 36 | MLBA-01-34 | West Laguna del Sello | −46.91 | 288.60 | 0.37 | 1.700 ± 0.020 | 725–1300 | 100.0 | 1.710 ± 0.010 | 0.85 | 5 of 5 | 0.48 | 294.3 ± 1.7 | **1.720 ± 0.020** | T |
| PAT 38 | MLBA-01-36 | West Laguna del Sello | −46.97 | 288.58 | 0.72 | 1.740 ± 0.020 | 725–1260 | 100.0 | 1.730 ± 0.020 | 0.18 | 7 of 7 | 0.16 | 295.7 ± 0.7 | **1.730 ± 0.020** | T |
| PAT 53 | MLBA-01-51 | Cerro Lapiz section | −47.12 | 288.28 | 0.35 | 2.435 ± 0.023 | 850–1350 | 100.0 | 2.433 ± 0.021 | 0.45 | 5 of 5 | 0.41 | 298.0 ± 6.6 | **2.422 ± 0.035** | R |
| PAT 52 | MLBA-01-50 | Cerro Lapiz section bottom flow | −47.12 | 288.28 | 0.47 | 3.030 ± 0.030 | 700–1260 | 100.0 | 3.030± 0.020 | 0.50 | 5 of 5 | 0.60 | 294.9 ± 2.8 | **3.030 ± 0.030** | R |
| PAT 54 | MLBA-01-57 | Cerro Lapiz neck | −47.12 | 288.27 | 0.32 | 3.180 ± 0.050 | 700–1330 | 100.0 | 3.180 ± 0.030 | 0.46 | 7 of 7 | 0.47 | 295.0 ± 1.6 | **3.180 ± 0.040** | N |
| PAT 56 | MLBA-01-55 | Dacite dome 2648 m | −47.00 | 288.24 | 0.61 | 3.280 ± 0.040 | 1300–1675 | 78.2 | 3.230 ± 0.030 | 0.69 | 3 of 5 | 1.13 | 288.1 ± 30.3 | [d]**3.290 ± 0.220** | R |
| not drilled | MLBA-98-003d | Cerro Overo | −46.82 | 288.83 | 0.23 | 6.810 ± 0.060 | 825–1050 | 64.3 | 6.850 ± 0.060 | 0.03 | 3 of 6 | 0.01 | 294.3 ± 10.2 | **6.870 ± 0.210** | |
| not drilled | RIN-96-01 | NE MLBA Est. Rincon | −46.77 | 289.19 | 0.18 | 7.710 ± 0.070 | 725–1300 | 100.0 | 7.720 ± 0.070 | 0.61 | 5 of 5 | 0.70 | 298.1 ± 11.3 | **7.690 ± 0.160** | |
| PAT 32 | MLBA-01-31 | East Laguna del Sello | −46.90 | 288.90 | 0.46 | 7.660 ± 0.050 | 1000–1175 | 74.2 | 7.720 ± 0.050 | 1.97 | 4 of 7 | 3.00 | 296.3 ± 11.7 | **7.710 ± 0.070** | R |

**Table 2.** (continued)

| Paleomag Site # | Field ID # | Unit or Location | Lat. | Long. | K/Ca total | Total fusion Age (Ma) ± 2σ | Increments used, °C | Age Spectrum $^{39}$Ar% | Age (Ma) ± 2σ | MSWD | N | Isochron Analysis SUMS (N-2) | $^{40}$Ar/$^{36}$Ar$_i$ ± 2σ | [a]Age (Ma) ± 2σ | Polarity |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| PAT 41 | MLBA-01-38 | South Laguna del Sello | −46.90 | 288.70 | 0.31 | 7.990 ± 0.060 | 950–1080 | 70.2 | 8.040 ± 0.070 | 2.14 | 4 of 10 | 0.34 | 325.1 ± 25.0 | **7.860 ± 0.160** | T |
| not drilled | MLBA-01-58 | SE Corner of MLBA | −47.20 | 289.00 | 0.06 | 10.560 ± 0.450 | 800–1475 | 78.0 | 10.780 ± 0.610 | 1.44 | 4 of 5 | 0.76 | 297.5 ± 2.4 | **10.270 ± 0.790** | |
| PAT 51 | MLBA-01-49 | SE MLBA Vizcaino | −47.17 | 289.02 | 0.04 | 10.440 ± 0.380 | 930–1250 | 70.2 | 11.270 ± 0.390 | 1.19 | 7 of 11 | 0.81 | 298.4 ± 3.3 | **10.120 ± 1.350** | N |
| not drilled | MLBA-98-21 | NE MLBA Est. Telken | −46.82 | 289.09 | 0.07 | 11.280 ± 0.130 | 965–1430 | 84.7 | 11.040 ± 0.110 | 0.74 | 8 of 11 | 0.84 | 296.6 ± 6.5 | **11.030 ± 0.140** | |

[a] Ages calculated relative to 1.194 Ma Alder Creek Rhyolite sanidine [*Renne et al.*, 1998]; uncertainties reported at 2σ precision.
[b] Data from *Singer et al.* [2004].
[c] Data from *Ackert et al.* [2003]; mean age of two sites on Rio Pinturas flow is 0.066 ± 0.004 Ma.
[d] Analysis of matrix feldspar rather than basaltic groundmass.

Together with the $^{40}$Ar/$^{39}$Ar ages, these data indicate that 11 of the normally magnetized lavas erupted during the Brunhes chron, 17 erupted during the Matuyama chron, and 6 erupted during the Gauss or earlier chrons. However, not all the lavas temporally confined to the Matuyama chron are reversely magnetized. PAT 5, PAT 6, and PAT 1 preserve normal or transitional directions and ages corresponding to the Jaramillo normal subchron 1.07 to 0.99 Ma [e.g., *Singer et al.*, 1999; *Singer and Brown*, 2002]. Sites PAT 27 and PAT 57, that may correspond to eroded remnants of the single large Casa Piedra Flow (Figure 2), are ca. 1.25 to 1.29 Ma and transitionally magnetized, suggesting the possibility that this flow erupted during the onset of the Cobb Mountain event [e.g., *Singer et al.*, 1999]. Moreover, sites PAT 36 and PAT 38 are transitionally magnetized, suggesting that their ages of ca. 1.72 Ma may correspond to the termination of the Olduvai normal subchron. Ten other lavas erupted during the Matuyama chron are reversely magnetized.

## 8. Discussion

[26] The 36 paleomagnetic sites and the 38 new $^{40}$Ar/$^{39}$Ar dates from Meseta del Lago Buenos Aires provide important information from this here-to-for poorly studied region. Although the presence of late Tertiary basalts in Patagonia has been known for some time [*Mercer*, 1969; *Baker et al.*, 1981], this is the first detailed study of the geochronology and paleomagnetism of such rocks since the pioneering work of *Fleck et al.* [1972]. It adds to a growing body of petrologic and geochemical data that are used to refine models of slab window magmatism under southern South America [*Gorring and Kay*, 2001; *Gorring et al.*, 1997, 2003]. The paleomagnetic data allows us to investigate the time averaged field deep within the Southern Hemisphere as well as look at well-dated transitional results.

### 8.1. Transitional Directions

[27] Among the 36 paleomagnetic sites from MLBA are ten sites that have low latitude poles. As listed in Table 3, these sites have pole latitudes that range from 7.2° to 54.6°, putting them in the

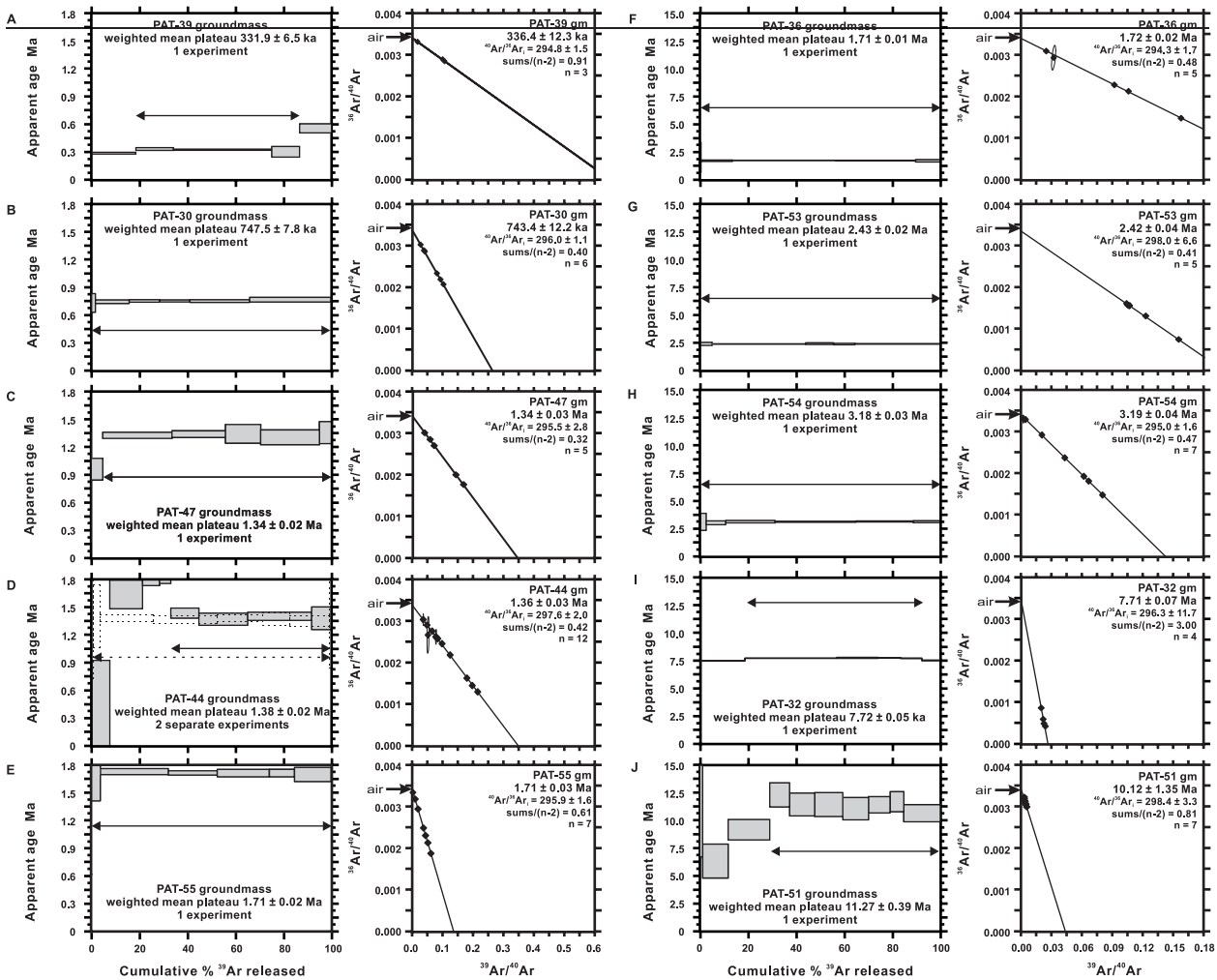



**Figure 8.** The $^{40}Ar/^{39}Ar$ age spectra and isochrons from 10 lavas representing the geographic and temporal coverage of the 38 lavas summarized in Table 2. Horizontal arrows indicate the plateau increments that were used to calculate the isochrons. The isochrons give the preferred age of each lava.

classification of transitional directions. That over 20% of the total sites in this study yield transitional directions may at first seem excessive. Early work by *Doell and Cox* [1972] looking at over 1000 lava flows from around the globe with ages less than 10 Ma, found only about 2% of directions fell into a transitional (low latitude pole) classification. Recent paleomagnetic and geochronologic results [e.g., *Singer et al.*, 2002] indicate that transitional directions, especially those related to excursions, are perhaps more common in the magnetic record than previously thought. *Constable and Johnson* [1999] looking at statistical models for paleosecular variation compared to lava data over the past 5 Ma find that many observables are anisotropic. This includes the percentage of VGP latitudes <45°, which for the Southern Hemisphere at the latitude of MLBA is 5% or 10% depending on the model used. Thus it seems possible that periods of transitional field behavior could be larger in the Southern Hemisphere, but still not as large as seen in this study. The question remains as to whether these sites are far from the expected directions due to other circumstances- such as lightning strikes, undetected local tectonics (block rotations, small-scale tilting, non-horizontal emplacement), or subsequent alteration of magnetic mineralogy, or whether they represent true field positions.

[28] Geochronologic data helps to investigate the validity of these transitional directions in that age determinations from all the sites are closely related

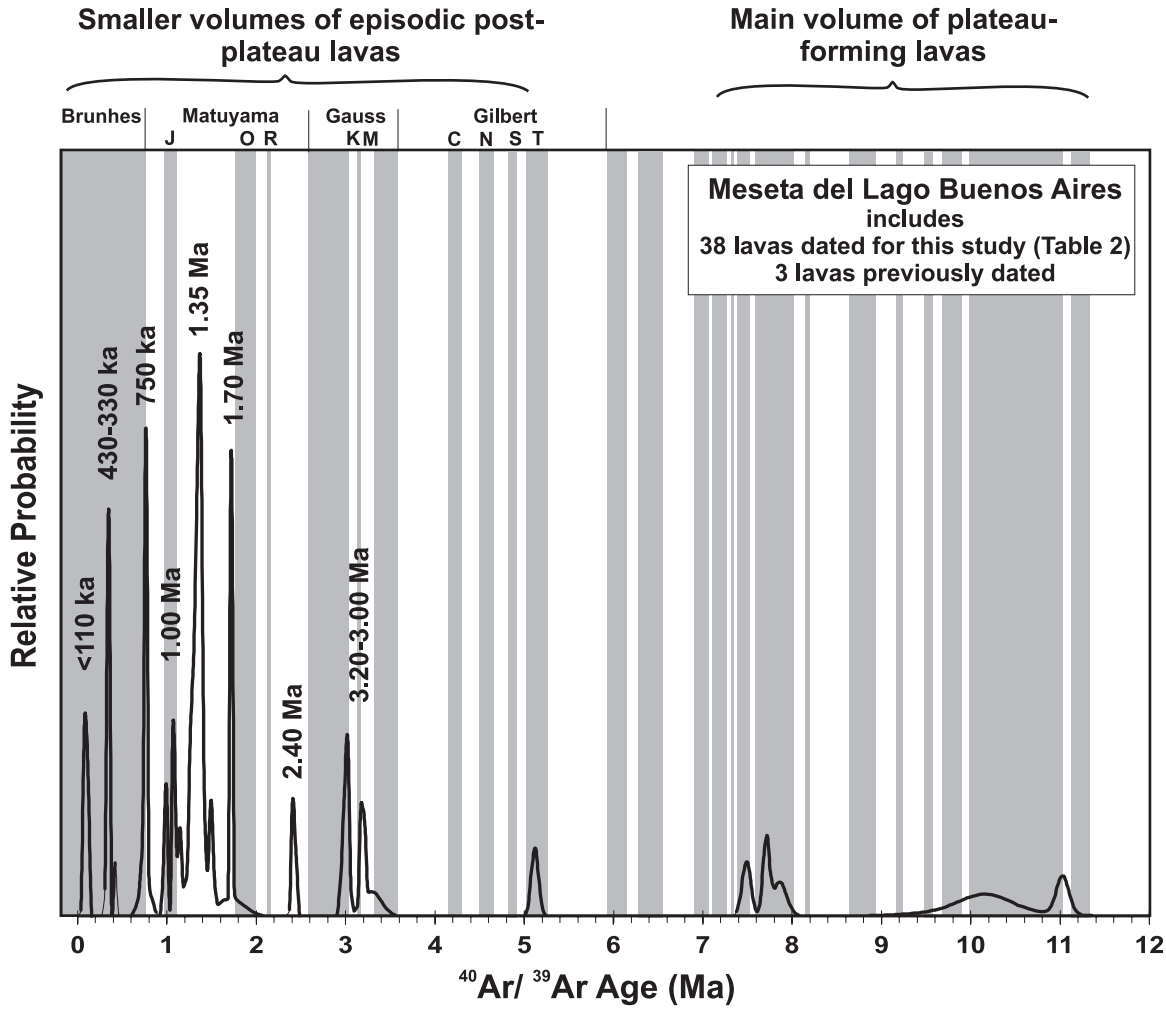



**Figure 9.** Relative probability diagram for the forty-one $^{40}Ar/^{39}Ar$-dated lavas in Figure 2. Thirty eight of the ages are from this study, whereas three are from *Ton-That et al.* [1999]. At least 11 periods of peak activity are distinguishable, with an age distribution suggesting that most of the lavas erupted during the late Matuyama and Brunhes chrons. Grey vertical bands denote normal polarity intervals (modified from *Baski* [1995] and *Berggren et al.* [1995] so that chron boundaries are consistent with the values of the neutron fluence monitors used in this study).

to known chron or subchron boundaries. As indicated in Table 3, the ten transitional directions fall into several groups. Two flows, PAT 6 at 0.984 ± 0.035 Ma and PAT 1, with an age of 1.016 ± 0.010 Ma, correspond to the end and the upper part of the Jaramillo subchron, currently defined as existing from 1.069 ± 0.006 Ma to 1.000 ± 0.005 Ma (*Singer et al.* [1999]; age of chron boundaries here and below were adjusted to reflect intercalibrated values of neutron fluence standards in *Renne et al.* [1998], that have been adopted in this paper). There are four sites (PAT 27, 28, 48 and 57) with ages between 1.25 and 1.32 Ma that have poles with low southern latitudes and longitudes near 300° (Table 3; Figure 7). These ages are slightly older than those currently associated with the Cobb Mountain Subchron of 1.194 ± 0.006 Ma [*Turrin et al.*, 1994; *Renne et al.*, 1998; *Singer et al.*, 2002], but may represent a short pre-Cobb Mountain feature observed in recent ODP cores from the Bermuda Rise [*Yang et al.*, 2001]. Two flows on the Meseta, PAT 36 and 38, have ages of 1.72 and 1.73 Ma respectively, as well as near-equatorial poles. These ages are slightly younger than the accepted age of the Olduvai Subchron (1.95 to 1.77 Ma; adjusted from *Cande and Kent* [1995]), but given the 2σ uncertainties, they are only 20 kyr older than the weighted mean of two K-

**Table 3.** Paleomagnetic Directions and Ages of Sites with Transitional Directions[a]

| Site | Age | Direction | Pole | Event |
|---|---|---|---|---|
| PAT 37 | 0.341 ± 0.033 | −80.7, 55.5 | 54.6, 82.3 | ??? |
| PAT 6 | 0.984 ± 0.035 | −84.8, 317.3 | 54.0, 121.2 | End of Jaramillo |
| PAT 1 | 1.016 ± 0.010 | −47.7, 183.6 | −14.0, 112.5 | During Jaramillo |
| PAT 57 | 1.250 ± 0.031 | 68.4, 32.2 | −12.5, 308.8 | Cobb Mountain |
| PAT 27 | 1.290 ± 0.030 | 84.6, 34.6 | −38.1, 296.8 | Or |
| PAT 28 | * | 77.6, 75.7 | −37.1, 318.4 | Pre-Cobb Mt |
| PAT 48 | 1.315 ± 0.029 | 81.6, 27.8 | −32.1, 298.0 | |
| PAT 36 | 1.720 ± 0.020 | 50.1, 279.5 | −16.1, 226.8 | End of Olduvai? or |
| PAT 38 | 1.730 ± 0.020 | 47.2, 321.9 | 7.2, 245.4 | Post-Olduvai |
| PAT 41 | 7.860 ± 0.160 | −16.6, 112.0 | −8.4, 40.8 | Onset of C4n.2n |

[a] Age, isochron age, in Ma, with ±2σ; DIRECTION, inclination and declination in degrees; POLE, latitude and longitude of virtual geomagnetic pole, in degrees. Asterisks indicates lava directly overlies PAT 27.

Ar ages of 1.62 ± 0.06 Ma from basalt flows thought to record the elusive Gilsa event [*Udagawa et al.*, 1999]. Finally, the oldest transitional direction, PAT 41, dated at 7.860 ± 0.16 Ma, is coincident with that of the termination of a normal chron, C4n.2n, at 7.76 Ma (adjusted from *Cande and Kent* [1995]). Only the youngest flow, PAT 37 at 0.341 ± 0.033 Ma is not readily correlated with a known polarity event or excursion.

[29] In addition to the consistent agreement of the geochronology with known periods of field transition, the magnetic data supports the contention that true field behavior is being measured. Although none of these sites exhibit low NRM intensity one might associate with transitional field positions, they also do not have high intensities that might be associated with lightning strikes (Figure 4b). Average NRM for these 10 flows is 4.68 A/m, only slightly greater than the average for the entire set of 4.03 A/m. These sites do have a larger average $\alpha_{95}$ than the total group (6.3°Compared to 4.9°), but with 8 of the 10 flows having $\alpha_{95} \leq 6.6°$ (Figure 4c). The data from these flows hardly classify as ''scattered'' but rather are only slightly more dispersed than the acceptable flows. The fact that there are several flows with similar ages and similar directions (Table 3), but not co-located, rules out local tilting or small block rotations as a cause of aberrant directions. The transitional site with the greatest scatter (PAT 38 with $\alpha_{95} = 13.9°$) has an identical age to the transitional flow with the least scatter (PAT 36, with $\alpha_{95} = 2.6°$). It appears that these sites, with some caution, can be considered to represent field behavior at the time of their emplacement.

## 8.2 Paleosecular Variation

[30] Recently there has been renewed interest in providing PSV information from young volcanic rocks [e.g., *Tauxe et al.*, 2000, 2003; *Camps et al.*, 2001; *Herrero-Bervera and Valet*, 2003; *Mejia et al.*, 2002], as well as attempts to improve old data sets [*Coe et al.*, 2000; *Brown*, 2002]. Much of this renewed interest is a result of the need to have excellent time averaged field data from a number of wide-spread locations, including high latitudes in the Northern Hemisphere and more data from any latitude in the Southern Hemisphere. These spot observations of the magnetic field are then combined into a global database, such as the ones presented by *Quidelleur et al.* [1994] and *Johnson and Constable* [1996] and used by them and others with respect to the TAF[e.g., *Johnson and Constable*, 1997; *Constable and Johnson*, 1999; *Kelly and Gubbins*, 1997; *Hatakeyama and Kono*, 2002].

[31] One of several ways to view paleosecular variation at any one location is to investigate the dispersion of directions or poles. It has long been common practice to calculate the dispersion of the VGPs about the rotation axis ($S_p$) and then compare this value to predicted models [e.g., *Doell and Cox*, 1972]. Table 4 reports the angular dispersion of the VGPs from Meseta del Lago Buenos Aries for the entire collection as well as the two subsets discussed above, after correcting for within-site dispersion using the method of *McElhinny and McFadden*

**Table 4.** Paleosecular Variation Statistics for Meseta del Lago Buenos Aires Lava Flows[a]

| Group | Number | Direction | k | $\alpha_{95}$ | Pole | $A_{95}$ | $S_F$ | $S_u$ | $S_l$ |
|---|---|---|---|---|---|---|---|---|---|
| All sites | 36 | −72.8, 12.9 | 9 | 8.3 | 85.1, 118.8 | 12.0 | 38.5 | 45.9 | 33. |
| Selection I | 26 | −63.0, 3.4 | 29 | 5.4 | 87.7, 8.7 | 7.3 | 20.0 | 24.8 | 16.8 |
| Normal | 14 | −64.9, 7.7 | 30 | 7.4 | 84.4, 41.2 | | | | |
| Reversed | 12 | 60.7, 179.1 | 27 | 8.5 | −84.7, 101.3 | | | | |
| Selection II | 30 | −68.5, 11.4 | 10 | 8.7 | 88.4, 45.5 | 11.3 | 33.5 | 40.6 | 28.5 |

[a] Number, number of sites in group; Direction, mean inclination and declination of group; K, precision parameter; $\alpha_{95}$ and $A_{95}$, 95% confidence circle about mean direction and mean pole; Pole, mean latitude and longitude of group pole; $S_F$, dispersion of poles with respect to geographic pole corrected for within-site dispersion; $S_u$ and $S_l$, upper and lower 95% limits on $S_F$ from *Cox* [1969]. Selection I and II described in text.

[1997]. Both the total population and Selection II yield large values for dispersion of the field ($S_f$), while the subset of flows with pole latitudes > 55° gives a dispersion of 20.0° with 95% error limits of 16.8° and 24.8°. This value is plotted on Figure 10 for comparison to the fit of paleosecular variation data from lava flows for the last 5 million years to Model G of *McFadden et al.* [1991]. The data from MLBA Selection I lies appreciably above Model G, even with 95% error limits, indicating this data is more dispersed than predicted from Model G.

[32] To consider whether the dispersion viewed here is a valid measure of the TAF in South America requires other data sets use for comparison. Current work on lavas from the Pali Aike field in southern Patagonia (52°) yields a dispersion value closer to Model G [*Mejia et al.*, 2001]. Work on lavas from Tatara-San Pedro volcanic complex in central Chile (36°S) has produced a large number of measurements on lava flows younger than the Brunhes/Matuyama reversal (Brown, unpublished data, 2002). Initial results indicate rather large dispersion (18.2°) but additional flow statistics must be incorporated to obtain final results. Other data from the Southern Hemisphere includes results from the Crozet Archipelago [*Camps et al.*, 2001], also at 47°S. Results of 45 lava flows, also using a 45° latitude cut-off, give angular dispersion of 18.6° with errors of 14.6° and 22.4°, showing good overlap with the MLBA results. Results from Easter Island [*Brown*, 2002] at 27°S lie right on Model G, with a dispersion of 14.1° and error limits of 12.4° and 16.2°.

[33] At this point it is difficult to know if the higher than expected dispersion in southern South America is due to actual field behavior in this part of the world, or whether it is an artifact of inadequate or biased sampling. *Tauxe et al.* [2003] make a case for the TAF only being accurately determined from data sets of several hundred flows. Working in the American southwest, they combine studies over an area of 5° latitude and nearly 20° longitude. If this is the case, areas such as southern South America will need many more studies to provide the necessary data to define the TAF.

## 9. Conclusions

[34] To augment and improve observations relevant to understanding the time averaged character of the Earth's magnetic field, we have measured paleomagnetic directions and determined $^{40}Ar/^{39}Ar$ ages from three dozen lava flows or shallow intrusions at Meseta del Lago Buenos Aires, 47°S in Argentine Patagonia. Our results lead to the following conclusions:

[35] 1. Thirty-six lava flows give stable paleomagnetic directions. After omitting sites with low latitude poles (<55°), the remaining 26 flows have a mean direction of inclination = −63.0°, declination = 3.4°, $\alpha_{95}$ = 5.4°.

[36] 2. Thirty-eight new $^{40}Ar/^{39}Ar$ incremental heating experiments on all but two of the paleomagnetic sites gave ages that range from 11.030 ± 0.140 Ma to 0.067 ± 0.004 Ma, with the bulk of the ages being younger than 3.3 Ma.

[37] 3. Geochronology indicates that the volcanism was episodic with peak periods of eruptive activity at 3.20–3.00 Ma, 2.40 Ma, 1.70 Ma. 1.35 Ma, 1.00 Ma, 750 ka, 430–330 ka and <110 ka.

[38] 4. Transitional flows, as determined by shallow VGP latitudes, make up over 20% of the total population. The ages determined from the transi-

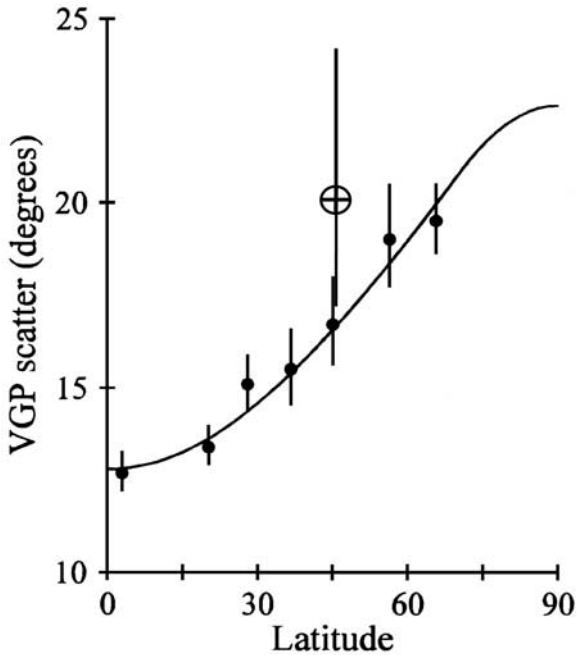



**Figure 10.** Least squares fit of Model G to both normal and reversed polarity scatter of virtual geomagnetic poles from lavas for the past 5 million years, from *McFadden et al.* [1991]. Open circle is VGP dispersion for the selected sites (N = 26) from Meseta del Lago Buenos Aires, shown with 95% confidence limits after *Cox* [1969].

tional flows indicate they were emplaced within or near short subchrons or at reversal boundaries.

[39] 5. Secular variation of Selection I lava flows gives an angular dispersion of VGPs of 20.0°, larger than dispersion predicted by Model G. It is not known if this larger dispersion is in response to regional variations in the magnetic field or is an artifact of inadequate sampling.

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